

What is Pandas

Pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language.

<https://pandas.pydata.org/about/index.html>

Pandas Series

A Pandas Series is like a column in a table. It is a 1-D array holding data of any type.

✖ Importing Pandas

```
import numpy as np
import pandas as pd
```

✖ Series from lists

```
# string
country = ['India', 'Pakistan', 'USA', 'Nepal', 'Srilanka']

pd.Series(country)
```

```
0      India
1    Pakistan
2        USA
3      Nepal
4    Srilanka
dtype: object
```

```
# integers
runs = [13,24,56,78,100]

runs_ser = pd.Series(runs)
```

```
# custom index
marks = [67,57,89,100]
subjects = ['maths', 'english', 'science', 'hindi']

pd.Series(marks, index=subjects)
```

```
maths      67
english    57
science    89
hindi     100
dtype: int64
```

```
# setting a name
marks = pd.Series(marks, index=subjects, name='Nitish ke marks')
marks
```

```
maths      67
english    57
science    89
hindi     100
Name: Nitish ke marks, dtype: int64
```

✖ Series from dict

```
marks = {
    'maths':67,
    'english':57,
    'science':89,
    'hindi':100
}
```

```
marks_series = pd.Series(marks,name='nitish ke marks')
marks_series
```

```
maths      67
english    57
science    89
hindi     100
Name: nitish ke marks, dtype: int64
```

Series Attributes

```
# size
marks_series.size
```

```
4
```

```
# dtype
marks_series.dtype
```

```
dtype('int64')
```

```
# name
marks_series.name
```

```
'nitish ke marks'
```

```
# is_unique
marks_series.is_unique
```

```
pd.Series([1,1,2,3,4,5]).is_unique
```

```
False
```

```
# index
marks_series.index
```

```
Index(['maths', 'english', 'science', 'hindi'], dtype='object')
```

```
runs_ser.index
```

```
RangeIndex(start=0, stop=5, step=1)
```

```
# values
marks_series.values
```

```
array([ 67,  57,  89, 100])
```

Series using read_csv

```
# with one col
subs = pd.read_csv('/content/subs.csv',squeeze=True)
subs
```

```
0      48
1      57
2      40
3      43
4      44
...
360    231
361    226
362    155
363    144
364    172
Name: Subscribers gained, Length: 365, dtype: int64
```

```
# with 2 cols
vk = pd.read_csv('/content/kohli_ipl.csv',index_col='match_no',squeeze=True)
vk
```

```
match_no
1         1
```

```

2    23
3    13
4    12
5     1
..
211   0
212   20
213   73
214   25
215    7
Name: runs, Length: 215, dtype: int64

```

```

movies = pd.read_csv('/content/bollywood.csv', index_col='movie', squeeze=True)
movies

```

```

movie
Uri: The Surgical Strike          Vicky Kaushal
Battalion 609                    Vicky Ahuja
The Accidental Prime Minister (film)  Anupam Kher
Why Cheat India                  Emraan Hashmi
Evening Shadows                  Mona Ambegaonkar
...
Hum Tumhare Hain Sanam          Shah Rukh Khan
Aankhen (2002 film)             Amitabh Bachchan
Saathiya (film)                 Vivek Oberoi
Company (film)                  Ajay Devgn
Awara Paagal Deewana            Akshay Kumar
Name: lead, Length: 1500, dtype: object

```

Series methods

```

# head and tail
subs.head()

```

```

0    48
1    57
2    40
3    43
4    44
Name: Subscribers gained, dtype: int64

```

```
vk.head(3)
```

```

match_no
1     1
2    23
3    13
Name: runs, dtype: int64

```

```
vk.tail(10)
```

```

match_no
206     0
207     0
208     9
209    58
210    30
211     0
212    20
213    73
214    25
215     7
Name: runs, dtype: int64

```

```

# sample
movies.sample(5)

```

```

movie
Arjun: The Warrior Prince      Yudhveer Bakoliya
Viceroy's House (film)        Hugh Bonneville
Joggers' Park (film)          Victor Banerjee
Tere Mere Phere                Vinay Pathak
Mission Mangal                 Akshay Kumar
Name: lead, dtype: object

```

```

# value_counts -> movies
movies.value_counts()

```

```

Akshay Kumar      48
Amitabh Bachchan  45
Ajay Devgn        38
Salman Khan       31
Sanjay Dutt       26
..
Diganth           1
Parveen Kaur      1
Seema Azmi        1
Akanksha Puri     1
Edwin Fernandes   1
Name: lead, Length: 566, dtype: int64

```

```

# sort_values -> inplace
vk.sort_values(ascending=False).head(1).values[0]

```

```
113
```

```
vk.sort_values(ascending=False)
```

```

match_no
128    113
126    109
123    108
164    100
120    100
...
93      0
211     0
130     0
8        0
135     0
Name: runs, Length: 215, dtype: int64

```

```

# sort_index -> inplace -> movies
movies.sort_index(ascending=False,inplace=True)

```

```
movies
```

```

movie
Zor Lagaa Ke...Haiya!      Meghan Jadhav
Zokkomon                   Darsheel Safary
Zindagi Tere Naam          Mithun Chakraborty
Zindagi Na Milegi Dobara   Hrithik Roshan
Zindagi 50-50              Veena Malik
...
2 States (2014 film)       Anjun Kapoor
1971 (2007 film)          Manoj Bajpayee
1920: The Evil Returns     Vicky Ahuja
1920: London               Sharman Joshi
1920 (film)                Rajnesh Duggall
Name: lead, Length: 1500, dtype: object

```

```
vk.sort_values(inplace=True)
```

```
vk
```

```

match_no
87      0
211     0
207     0
206     0
91      0
...
164    100
120    100
123    108
126    109
128    113
Name: runs, Length: 215, dtype: int64

```

Series Maths Methods

```

# count
vk.count()

```

215

```
# sum -> product
subs.sum()
```

49510

```
# mean -> median -> mode -> std -> var
subs.mean()
print(vk.median())
print(movies.mode())
print(subs.std())
print(vk.var())
```

```
24.0
0    Akshay Kumar
dtype: object
62.6750230372527
688.0024777222343
```

```
# min/max
subs.max()
```

396

```
# describe
subs.describe()
```

```
count    365.000000
mean     135.643836
std       62.675023
min       33.000000
25%       88.000000
50%      123.000000
75%      177.000000
max       396.000000
Name: Subscribers gained, dtype: float64
```

▼ Series Indexing

```
# integer indexing
x = pd.Series([12,13,14,35,46,57,58,79,9])
x
```

```
0    12
1    13
2    14
3    35
4    46
5    57
6    58
7    79
8     9
dtype: int64
```

```
# negative indexing
x[-1]
```

```

-----
ValueError                                Traceback (most recent call last)
/usr/local/lib/python3.8/dist-packages/pandas/core/indexes/range.py in get_loc(self, key, method, tolerance)
    384         try:
--> 385             return self._range.index(new_key)
    386         except ValueError as err:

```

ValueError: -1 is not in range

The above exception was the direct cause of the following exception:

```

KeyError                                Traceback (most recent call last)
<ipython-input-103-60055cb99cf0> in <module>
      1 # negative indexing
----> 2 x[-1]

-----
      2 frames -----
/usr/local/lib/python3.8/dist-packages/pandas/core/indexes/range.py in get_loc(self, key, method, tolerance)
    385         return self._range.index(new_key)
    386         except ValueError as err:
--> 387             raise KeyError(key) from err
    388         raise KeyError(key)
    389         return super().get_loc(key, method=method, tolerance=tolerance)

```

KeyError: -1

movies

```

movie
Zor Lagaa Ke...Haiya!      Meghan Jadhav
Zokkomon                  Darsheel Safary
Zindagi Tere Naam          Mithun Chakraborty
Zindagi Na Milegi Dobara   Hrithik Roshan
Zindagi 50-50              Veena Malik
...
2 States (2014 film)       Arjun Kapoor
1971 (2007 film)           Manoj Bajpayee
1920: The Evil Returns     Vicky Ahuja
1920: London               Sharman Joshi
1920 (film)                Rajnesh Duggall
Name: lead, Length: 1500, dtype: object

```

vk[-1]

```

-----
KeyError                                Traceback (most recent call last)
/usr/local/lib/python3.8/dist-packages/pandas/core/indexes/base.py in get_loc(self, key, method, tolerance)
    3360         try:
-> 3361             return self._engine.get_loc(casted_key)
    3362         except KeyError as err:

```

```

-----
      4 frames -----
pandas/_libs/hashtable_class_helper.pxi in pandas._libs.hashtable.Int64HashTable.get_item()
pandas/_libs/hashtable_class_helper.pxi in pandas._libs.hashtable.Int64HashTable.get_item()

```

KeyError: -1

The above exception was the direct cause of the following exception:

```

KeyError                                Traceback (most recent call last)
<ipython-input-107-8ecb27b68523> in <module>
----> 1 vk[-1]

/usr/local/lib/python3.8/dist-packages/pandas/core/indexes/base.py in get_loc(self, key, method, tolerance)
    3361         return self._engine.get_loc(casted_key)
    3362         except KeyError as err:
-> 3363             raise KeyError(key) from err
    3364
    3365         if is_scalar(key) and isna(key) and not self.hasnans:

```

KeyError: -1

marks_series[-1]

100

```
# slicing
vk[5:16]
```

```
match_no
6      9
7     34
8      0
9     21
10     3
11    10
12    38
13     3
14    11
15    50
16     2
Name: runs, dtype: int64
```

```
# negative slicing
vk[-5:]
```

```
match_no
211    0
212   20
213   73
214   25
215    7
Name: runs, dtype: int64
```

```
movies[:,2]
```

```
movie
Zor Lagaa Ke...Haiya!      Meghan Jadhav
Zindagi Tere Naam          Mithun Chakraborty
Zindagi 50-50              Veena Malik
Zinda (film)               Sanjay Dutt
Zid (2014 film)            Mannara Chopra
...
3 Storeys                  Aisha Ahmed
3 Deewarein                Naseeruddin Shah
22 Yards                   Barun Sobti
1971 (2007 film)           Manoj Bajpayee
1920: London               Sharman Joshi
Name: lead, Length: 750, dtype: object
```

```
# fancy indexing
vk[[1,3,4,5]]
```

```
match_no
1      1
3     13
4     12
5      1
Name: runs, dtype: int64
```

```
# indexing with labels -> fancy indexing
movies['2 States (2014 film)']
```

```
'Arjun Kapoor'
```

✖ Editing Series

```
# using indexing
marks_series[1] = 100
marks_series
```

```
maths      67
english   100
science    89
hindi     100
Name: nitish ke marks, dtype: int64
```

```
# what if an index does not exist
marks_series['evs'] = 100
```

```
marks_series
```

```

maths      67
english    100
science     89
hindi      100
sst         90
evs        100
Name: nitish ke marks, dtype: int64

```

```

# slicing
runs_ser[2:4] = [100,100]
runs_ser

```

```

0      13
1      24
2     100
3     100
4     100
dtype: int64

```

```

# fancy indexing
runs_ser[[0,3,4]] = [0,0,0]
runs_ser

```

```

0      0
1     24
2     100
3      0
4      0
dtype: int64

```

```

# using index label
movies['2 States (2014 film)'] = 'Alia Bhatt'
movies

```

```

movie
Zor Lagaa Ke...Haiya!      Meghan Jadhav
Zokkomon                  Darsheel Safary
Zindagi Tere Naam          Mithun Chakraborty
Zindagi Na Milegi Dobara   Hrithik Roshan
Zindagi 50-50              Veena Malik
...
2 States (2014 film)       Alia Bhatt
1971 (2007 film)           Manoj Bajpayee
1920: The Evil Returns     Vicky Ahuja
1920: London               Sharman Joshi
1920 (film)                Rajnesh Duggall
Name: lead, Length: 1500, dtype: object

```

▼ Copy and Views

Start coding or [generate](#) with AI.

▼ Series with Python Functionalities

```

# len/type/dir/sorted/max/min
print(len(subs))
print(type(subs))
print(dir(subs))
print(sorted(subs))
print(min(subs))
print(max(subs))

```

```

365
<class 'pandas.core.series.Series'>
['T', '_AXIS_LEN', '_AXIS_ORDERS', '_AXIS_REVERSED', '_AXIS_TO_AXIS_NUMBER', '_HANDLED_TYPES', '__abs__', '__add__', '__and__',
[33, 33, 35, 37, 39, 40, 40, 40, 40, 42, 42, 43, 44, 44, 44, 45, 46, 46, 48, 49, 49, 49, 49, 50, 50, 50, 51, 54, 56, 56, 56, 56,
33
396

```

```

# type conversion
list(marks_series)

```

```

[67, 100, 89, 100, 90, 100]

```



```
dict(marks_series)
```

```
{'maths': 67,
 'english': 100,
 'science': 89,
 'hindi': 100,
 'sst': 90,
 'evs': 100}
```

```
# membership operator
```

```
'2 States (2014 film)' in movies
```

```
True
```

```
'Alia Bhatt' in movies.values
```

```
True
```

```
movies
```

```
movie
Zor Lagaa Ke...Haiya!      Meghan Jadhav
Zokkomon                  Darsheel Safary
Zindagi Tere Naam          Mithun Chakraborty
Zindagi Na Milegi Dobara   Hrithik Roshan
Zindagi 50-50              Veena Malik
...
2 States (2014 film)       Alia Bhatt
1971 (2007 film)          Manoj Bajpayee
1920: The Evil Returns     Vicky Ahuja
1920: London              Sharman Joshi
1920 (film)               Rajnishes Duggall
Name: lead, Length: 1500, dtype: object
```

```
# looping
for i in movies.index:
    print(i)
```

```
Zor Lagaa Ke...Haiya!
Zokkomon
Zindagi Tere Naam
Zindagi Na Milegi Dobara
Zindagi 50-50
Zindaggi Rocks
Zinda (film)
Zila Ghaziabad
Zid (2014 film)
Zero (2018 film)
Zeher
Zed Plus
Zameer: The Fire Within
Zameen (2003 film)
Zamaanat
Yuvvraaj
Yuva
Yun Hota Toh Kya Hota
Youngistaan
Yeh Saali Aashiqui
Yeh Mera India
Yeh Lamhe Judaai Ke
Yeh Khula Aasmaan
Yeh Jawaani Hai Deewani
Yeh Hai India
Yeh Hai Bakrapur
Yeh Doorian
Yeh Dil
Yatra (2007 film)
Yamla Pagla Deewana: Phir Se
Yamla Pagla Deewana
Yakeen (2005 film)
Yadvi - The Dignified Princess
Yaaram (2019 film)
Ya Rab
Xcuse Me
Woodstock Villa
Woh Lamhe...
Why Cheat India
What's Your Raashee?
What the Fish
```

```

Well Done Abba
Welcome to Sajjanpur
Welcome Back (film)
Welcome 2 Karachi
Welcome (2007 film)
Wedding Pullav
Wedding Anniversary
Waris Shah: Ishq Daa Waaris
War Chhod Na Yaar
Waqt: The Race Against Time
Wanted (2009 film)
Wake Up Sid
Wake Up India
Wajah Tum Ho
Waiting (2015 film)
Waisa Bhi Hota Hai Part II
Wah Tai

```

```

# Arithmetic Operators(Broadcasting)
100 + marks_series

```

```

maths      167
english    200
science    189
hindi      200
sst        190
evs        200
Name: nitish ke marks, dtype: int64

```

```

# Relational Operators

```

```

vk >= 50

```

```

match_no
1      False
2      False
3      False
4      False
5      False
...
211     False
212     False
213      True
214     False
215     False
Name: runs, Length: 215, dtype: bool

```

Boolean Indexing on Series

```

# Find no of 50's and 100's scored by kohli
vk[vk >= 50].size

```

```

50

```

```

# find number of ducks
vk[vk == 0].size

```

```

9

```

```

# Count number of day when I had more than 200 subs a day
subs[subs > 200].size

```

```

59

```

```

# find actors who have done more than 20 movies
num_movies = movies.value_counts()
num_movies[num_movies > 20]

```

```

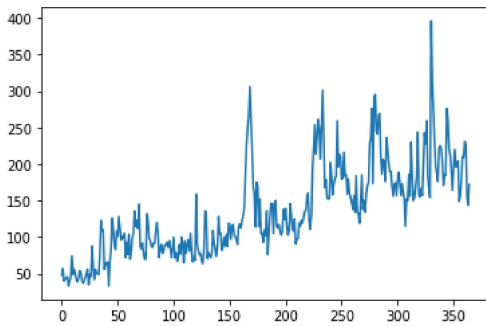
Akshay Kumar      48
Amitabh Bachchan   45
Ajay Devgn         38
Salman Khan        31
Sanjay Dutt        26
Shah Rukh Khan     22
Emraan Hashmi     21
Name: lead, dtype: int64

```

Plotting Graphs on Series

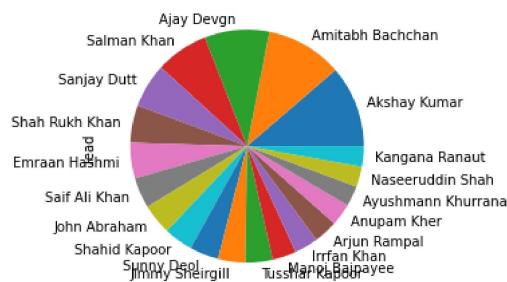
```
subs.plot()
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f54e0531a60>
```



```
movies.value_counts().head(20).plot(kind='pie')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x7f54e04f6850>
```



Some Important Series Methods

```
# astype
# between
# clip
# drop_duplicates
# isnull
# dropna
# fillna
# isin
# apply
# copy
```

```
import numpy as np
import pandas as pd
```

```
subs = pd.read_csv('/content/subs.csv', squeeze=True)
subs
```

```
0      48
1      57
2      40
3      43
4      44
...
360    231
361    226
362    155
363    144
364    172
Name: Subscribers gained, Length: 365, dtype: int64
```

```
vk = pd.read_csv('/content/kohli_ipl.csv', index_col='match_no', squeeze=True)
vk
```

```
match_no
1      1
```

```

2      23
3      13
4      12
5       1
..
211    0
212    20
213    73
214    25
215     7
Name: runs, Length: 215, dtype: int64

```

```

movies = pd.read_csv('/content/bollywood.csv',index_col='movie',squeeze=True)
movies

```

```

movie
Uri: The Surgical Strike          Vicky Kaushal
Battalion 609                    Vicky Ahuja
The Accidental Prime Minister (film)  Anupam Kher
Why Cheat India                  Emraan Hashmi
Evening Shadows                  Mona Ambegaonkar
...
Hum Tumhare Hain Sanam          Shah Rukh Khan
Aankhen (2002 film)             Amitabh Bachchan
Saathiya (film)                 Vivek Oberoi
Company (film)                  Ajay Devgn
Awara Paagal Deewana            Akshay Kumar
Name: lead, Length: 1500, dtype: object

```

```

# astype
import sys
sys.getsizeof(vk)

```

```
3456
```

```
sys.getsizeof(vk.astype('int16'))
```

```
2166
```

```

# between
vk[vk.between(51,99)].size

```

```
43
```

Start coding or [generate](#) with AI.

```

# clip
subs

```

```

0      48
1      57
2      40
3      43
4      44
...
360    231
361    226
362    155
363    144
364    172
Name: Subscribers gained, Length: 365, dtype: int64

```

```
subs.clip(100,200)
```

```

0      100
1      100
2      100
3      100
4      100
...
360    200
361    200
362    155
363    144
364    172
Name: Subscribers gained, Length: 365, dtype: int64

```

```
# drop_duplicates
temp = pd.Series([1,1,2,2,3,3,4,4])
temp
```

```
0    1
1    1
2    2
3    2
4    3
5    3
6    4
7    4
dtype: int64
```

```
temp.drop_duplicates(keep='last')
```

```
1    1
3    2
5    3
7    4
dtype: int64
```

```
temp.duplicated().sum()
```

```
4
```

```
vk.duplicated().sum()
```

```
137
```

```
movies.drop_duplicates()
```

```
movie
Uri: The Surgical Strike          Vicky Kaushal
Battalion 609                    Vicky Ahuja
The Accidental Prime Minister (film)  Anupam Kher
Why Cheat India                  Emraan Hashmi
Evening Shadows                  Mona Ambegaonkar
...
Sssshhh...                      Tanishaa Mukerji
Rules: Pyaar Ka Superhit Formula    Tanuja
Right Here Right Now (film)        Ankit
Talaash: The Hunt Begins...        Rakhee Gulzar
The Pink Mirror                    Edwin Fernandes
Name: lead, Length: 566, dtype: object
```

```
temp = pd.Series([1,2,3,np.nan,5,6,np.nan,8,np.nan,10])
temp
```

```
0    1.0
1    2.0
2    3.0
3    NaN
4    5.0
5    6.0
6    NaN
7    8.0
8    NaN
9   10.0
dtype: float64
```

```
temp.size
```

```
10
```

```
temp.count()
```

```
7
```

```
# isnull
temp.isnull().sum()
```

```
3
```

Start coding or [generate](#) with AI.

```
# dropna
temp.dropna()

0      1.0
1      2.0
2      3.0
4      5.0
5      6.0
7      8.0
9     10.0
dtype: float64
```

Start coding or generate with AI.

```
# fillna
temp.fillna(temp.mean())

0      1.0
1      2.0
2      3.0
3      5.0
4      5.0
5      6.0
6      5.0
7      8.0
8      5.0
9     10.0
dtype: float64
```

Start coding or generate with AI.

```
# isin
vk[(vk == 49) | (vk == 99)]

match_no
82      99
86      49
Name: runs, dtype: int64
```

```
vk[vk.isin([49,99])]
```

```
match_no
82      99
86      49
Name: runs, dtype: int64
```

Start coding or generate with AI.

```
# apply
movies
```

```
movie
Uri: The Surgical Strike          Vicky Kaushal
Battalion 609                    Vicky Ahuja
The Accidental Prime Minister (film)  Anupam Kher
Why Cheat India                  Emraan Hashmi
Evening Shadows                  Mona Ambegaonkar
...
Hum Tumhare Hain Sanam          Shah Rukh Khan
Aankhen (2002 film)             Amitabh Bachchan
Saathiya (film)                 Vivek Oberoi
Company (film)                  Ajay Devgn
Awara Paagal Deewana            Akshay Kumar
Name: lead, Length: 1500, dtype: object
```

```
movies.apply(lambda x:x.split()[0].upper())
```

```
movie
Uri: The Surgical Strike          VICKY
Battalion 609                    VICKY
The Accidental Prime Minister (film)  ANUPAM
Why Cheat India                  EMRAAN
Evening Shadows                  MONA
...
Hum Tumhare Hain Sanam          SHAH
Aankhen (2002 film)             AMITABH
Saathiya (film)                 VIVEK
```

```
Company (film)          AJAY
Awara Paagal Deewana    AKSHAY
Name: lead, Length: 1500, dtype: object
```

```
subs
```

```
0      48
1      57
2      40
3      43
4      44
...
360    231
361    226
362    155
363    144
364    172
Name: Subscribers gained, Length: 365, dtype: int64
```

```
subs.apply(lambda x: 'good day' if x > subs.mean() else 'bad day')
```

```
0      bad day
1      bad day
2      bad day
3      bad day
4      bad day
...
360    good day
361    good day
362    good day
363    good day
364    good day
Name: Subscribers gained, Length: 365, dtype: object
```

```
subs.mean()
```

```
135.64383561643837
```

```
# copy
```

```
vk
```

```
match_no
1      1
2     23
3     13
4     12
5      1
..
211    0
212    20
213    73
214    25
215     7
Name: runs, Length: 215, dtype: int64
```

```
new = vk.head()
```

```
new
```

```
match_no
1      1
2     23
3     13
4     12
5      1
Name: runs, dtype: int64
```

```
new[1] = 1
```

```
new = vk.head().copy()
```

```
new[1] = 100
```

new

match_no

1 100

2 23

3 13

4 12

5 1

Name: runs, dtype: int64

vk

match_no

1 1

2 23

3 13

4 12

5 1

..

211 0

212 20

213 73

214 25

215 7

Name: runs, Length: 215, dtype: int64